Name_____ Date_____ Period_____

LESSON 5-2 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Area of Triangles

Maryland Standard 6.G.A.1 · Learning target – I can find the area of a triangle using one half the base times the height.

REMEMBER

- A triangle is half of a parallelogram with the same base and height: **A = ½ × b × h**.
- The height meets the base at a right angle. It can fall **outside** the triangle.
- Area is in **square units** (cm², in², ft²).

WORKED EXAMPLE

Find the area of a triangle with base 14 cm and height 9 cm.

Step 1 Write the formula: A = ½ × b × h

Step 2 Substitute: A = ½ × 14 × 9

Step 3 Multiply: ½ × 14 = 7, and 7 × 9 = 63

Answer **63 cm²**

WORD BANK

base – the bottom side of the triangle

height – straight up from the base, at a square corner

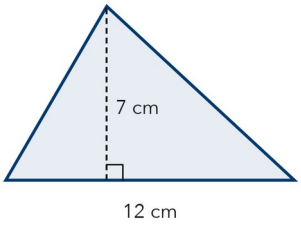
½ × b × h – half of base times height

area – space inside, in square units

PRACTICE

show your work · label units

1. Find the area of the triangle.



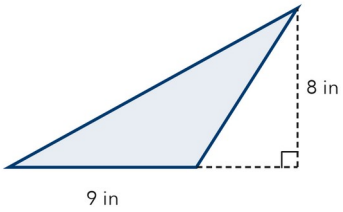
Step 1 Multiply the base and the height.

Step 2 Take half of that.

Show your work.

Answer_____ cm²

2. Find the area of the triangle. The height falls outside the triangle.



Step 1 Use 9 in and 8 in. The dashed line is the height.

Step 2 Multiply, then take half.

Show your work.

Answer_____ in²

3. Find the area of each triangle with base b and height h.

Step 1 Multiply b × h for each one.

Step 2 Take half.

a. b = 10 cm, h = 9 cm _____ cm²

b. b = 5 ft, h = 11 ft _____ ft²

4. Garden Bed. Maya's garden bed is a **triangle**: base 15 feet, height 6 feet. One bag of soil covers 9 square feet. How many bags does she need?

Step 1 Find the area: ½ × 15 × 6.

Step 2 Divide the area by 9.

Show your work.

Answer_____ bags

5. Find the Error. Jon says the area of a triangle with base 8 m and height 5 m is 40 m². Explain his mistake and find the correct area.

Frame *Jon forgot to ___. The area is ½ × 8 × 5 = ___ m².*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Find a triangle at home (a sign, a sandwich half, a roof). Estimate its base and height and its area.

Name_____ Date_____ Period_____

LESSON 5-2 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Missing Measures in Triangles

Maryland Standard 6.G.A.1 · Learning target – I can find a missing base or height of a triangle when I know its area.

REMEMBER

- Know the area and one measure? **Double the area**, then divide by the measure you know: **h = 2A ÷ b** and **b = 2A ÷ h**.
- Check: ½ × b × h should equal the given area.

WORKED EXAMPLE

Area = 30 in², base = 12 in. Find the height.

Step 1 Double the area: 2 × 30 = 60

Step 2 Divide by the base: 60 ÷ 12 = 5

Step 3 Check: ½ × 12 × 5 = 30, so it checks

Answer **h = 5 in**

WORD BANK

missing measure – the base or height you need to find

double – multiply by 2

divide – share into equal groups; undo multiplying

check – put your answer back in ½ × b × h

PRACTICE

show your work · label units

1. Complete the table. Each row describes one triangle.

Step 1 Double the area.

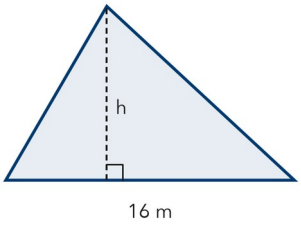
Step 2 Divide by the measure you know.

	Base	Height	Area
Triangle A	8 in		28 in²
Triangle B		9 ft	54 ft²

2. The area of the triangle is 48 m². Find the height *h*.

Step 1 Double the area: 2 × 48.

Step 2 Divide by the base, 16.



Show your work.

Answer_____ m

3. Find the missing measure of each triangle.

a. A = 40 m², b = 10 m; h = _____ m **b.** A = 27 in², h = 6 in; b = _____ in

4. Sail. A **sail** is a triangle. Its area is 35 square feet. Its height is 10 feet. How long is its base?

Step 1 Double the area: 2 × 35.

Step 2 Divide by 10.

Answer_____ ft

5. Compare. Triangle P: base 12 cm, height 5 cm. Triangle Q: base 6 cm, height 10 cm. Which has more area? Explain.

Frame *P has ___ cm² and Q has ___ cm², so ___.*

_____

_____

TEST PRACTICE

choose the best answer

6. A triangle has an area of 24 square units and a height of 4 units. What is the length of its base?

Step 1 Double 24.

Step 2 Divide by 4.

A 3 units **B** 6 units **C** 12 units **D** 48 units

7. Which expression gives the area of a triangle with base 18 m and height 5 m?

A 18 × 5 **B** ½ × 18 × 5 **C** 18 + 5 **D** 2 × 18 × 5

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___